Power converters

文件: power.kicad_sch

Basic-library composite additions

文件: basic-composites.kicad_sch

Panel and temperature

文件: panel.kicad_sch

Host interface and test points

文件: interface.kicad_sch

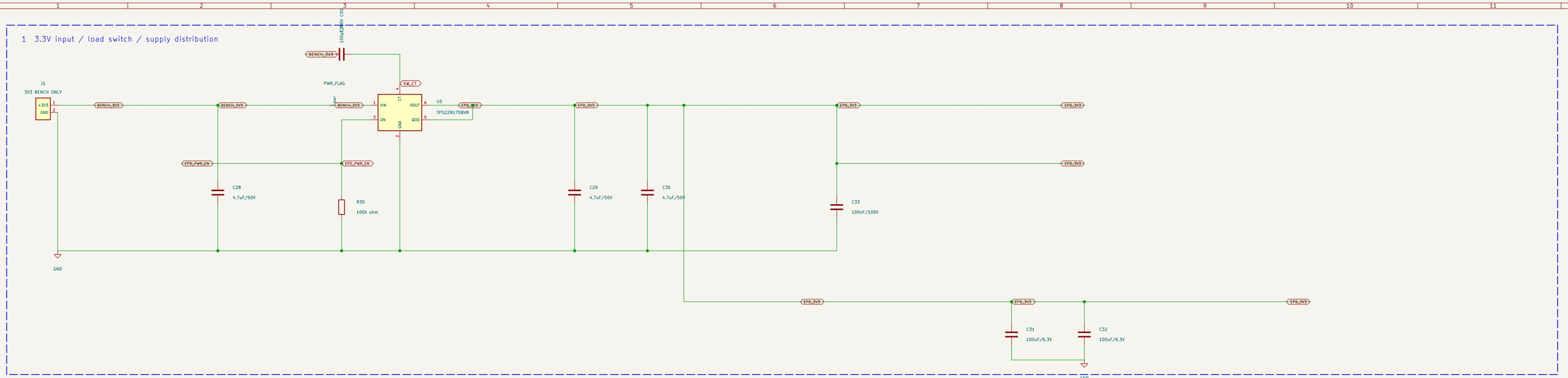
album

Sheet: /
File: gdep133c02-driver.kicad_sch

Title: GDEP133C02 minimum driver - DRAFT

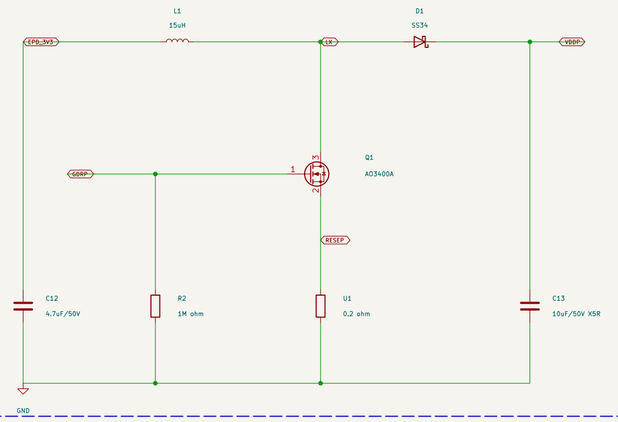
Size: A4	Date: 2026-10-09	**Rev: 0.11-BASIC17**
KiCad E.D.A. 10.0.6		Id: 1/5

1 3.3V input / load switch / supply distribution

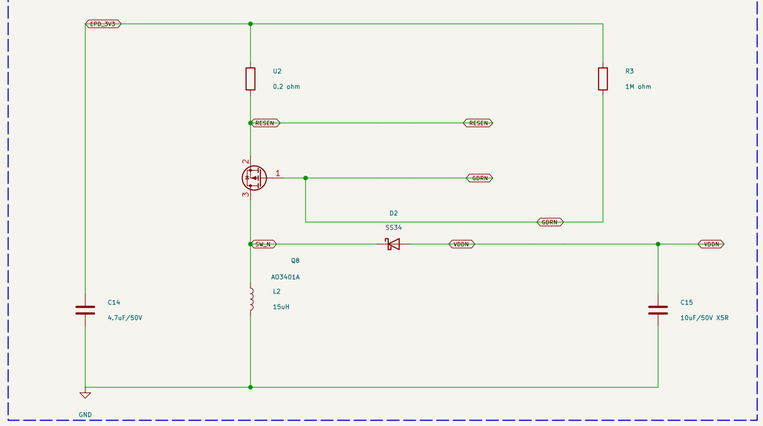


v0.4: U5 output = EPD_3V3; direct copper feeds to logic and power branches

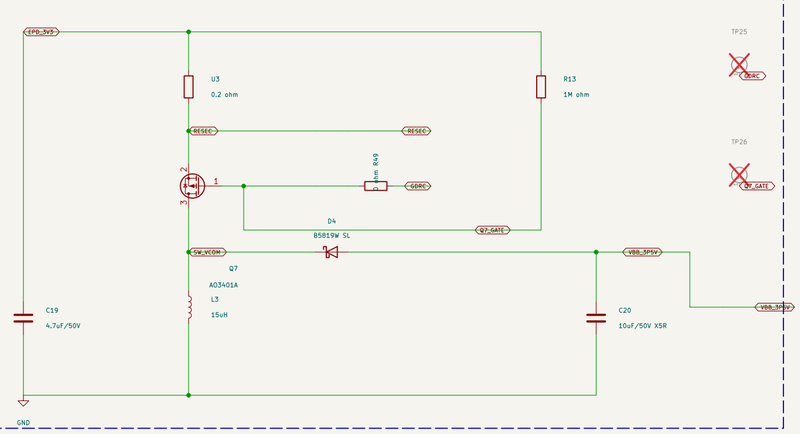
2 Positive source rail (VDDP)



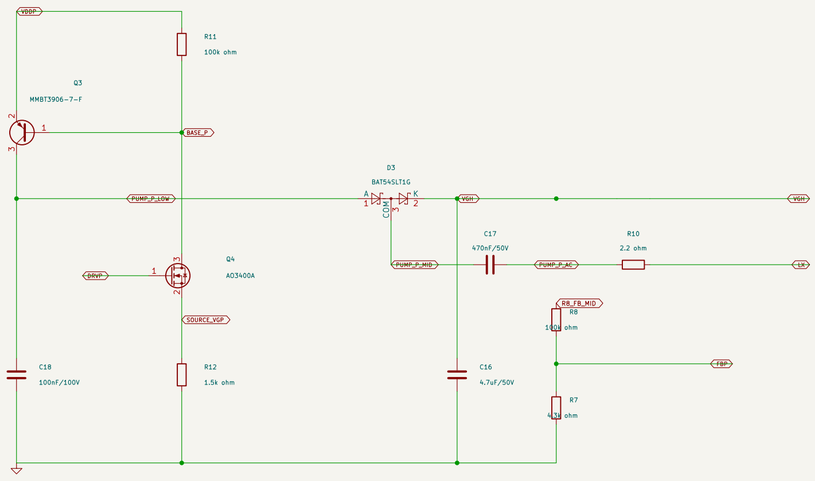
3 Negative source rail (VDDN)



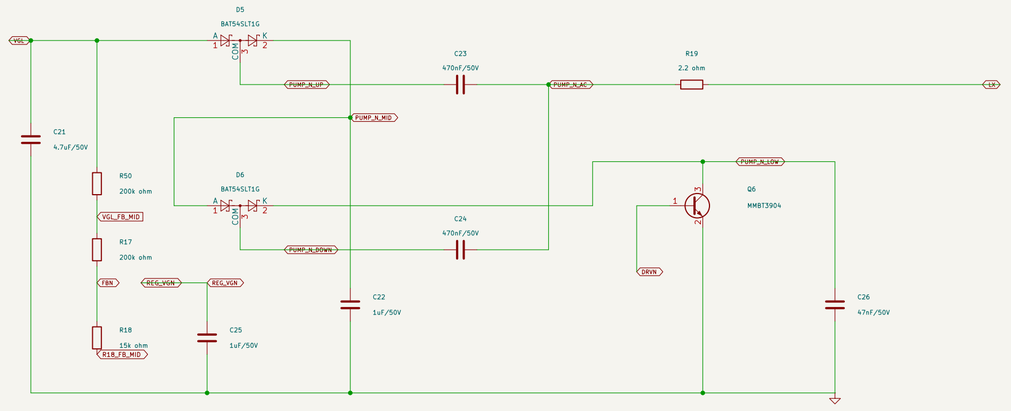
4 VCOM supply branch



5 Positive gate charge pump (VGH)



6 Negative gate charge pump (VGL)



Dots join wires; crossings without dots are not connected. Net labels retain the existing PCB net names.

Sheet: /Power converters/	
File: power.kicad_sch	
Title: Power – wired functional circuits	
Size: User	Date: 2026-10-09
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	Id: 2/5

1 Screen connector — pin numbers retained; functional pin layout

FPC1

FPC-05FB-60PH20

POWER

LOGIC / SPI

GATE / SOURCE

SOURCE RAILS

EPD_3V3	9	AVDD	
EPD_3V3	10	VDD	
EPD_3V3	19	VDDIO	
VCC	12	VCC	
	18	VCC1	
VDDP	13	VDDP	
	14	VDDP	
VDDN	16	VDDN	
	17	VDDN	
VGH	59	VGH	
VGL	60	VGL	
VBB_3P5V	41	VNCP_3P5V	
VBB_3P5V	56	VBB_3P5V	
VCOMBD_M	1	VCOMBD_M	
VCOMBD_S	36	VCOMBD_S	
RESEC	2	RESEC	
GDRC	3	GDRC	
RESEN	4	RESEN	
GDRN	5	GDRN	
RESEP	6	RESEP	
GDRP	7	GDRP	
DRVP	33	DRVP	
FBP	34	FBP	
DRVN	37	DRVN	
FBN	38	FBN	
REG_VGN	39	REG_VGN	
	8	GND	
	26	GND	
	35	GND	
	11	NC	
	15	NC	
	40	DRVN2	
	57	NC	
TSCL	20		
TSDA	21		
BS0	22		
BS1	23		
RES_N	24		
BUSY_N	25		
CS_M_N	27		
SCLK	28		
SI0	29		
SI1	30		
SI2	31		
SI3	32		
CS_S_N	58		
FPL_VCOM	54		
TFT_VCOM	55		
VSPH	42		
VSPH	45		
VSPL	43		
VSPL	46		
VSPL2	44		
VSPL2	47		
VSNH	48		
VSNH	53		
VSNL	49		
VSNL	52		
VSNL2	50		
VSNL2	51		

C7 10uF/50V X5R

C10 470nF/50V

C11 470nF/50V

2 SPI mode / signal defaults

R20 4.7k ohm

R21 4.7k ohm

R1 2k ohm

R35 10k ohm

C9 470nF/50V

R36 100k ohm

C8 4.7uF/50V

3 Panel-controlled temperature sensor

U4

TCN75AVDA713

C27 100nF/100V

R23 DNP: alert output is not strapped to ground.

4 ~~Source outputs / input straps / capacitor banks~~

C1	C103	C104	C105
C2	C106	C107	C108
C3	C109	C110	C111
C4	C112	C113	C114
C5	C115	C116	C117
C6	C118	C119	C120

All: 10uF/50V X5R; Nominal 33.3uF; effective capacitance pending.

Symbol grouping does not represent physical FPC pin positions.

Dots join wires; crossings without dots do not connect. Global labels are cross-sheet interfaces.

Sheet: /Panel and temperature/

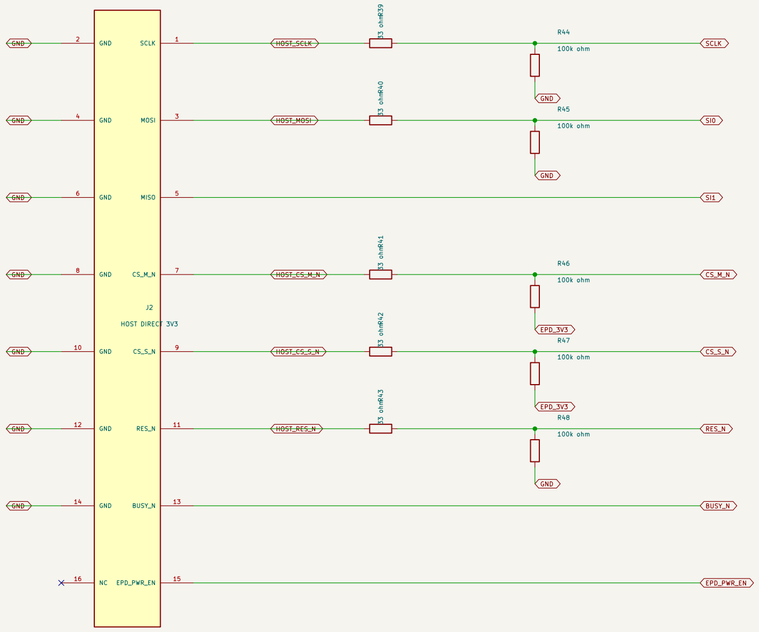
File: panel.kicad_sch

Title: Panel — wired functional circuits

Size: User | Date: 2026-10-09 | **Rev: 0.11-BASIC17**

KiCad E.D.A. 10.0.6 | Id: 3/5

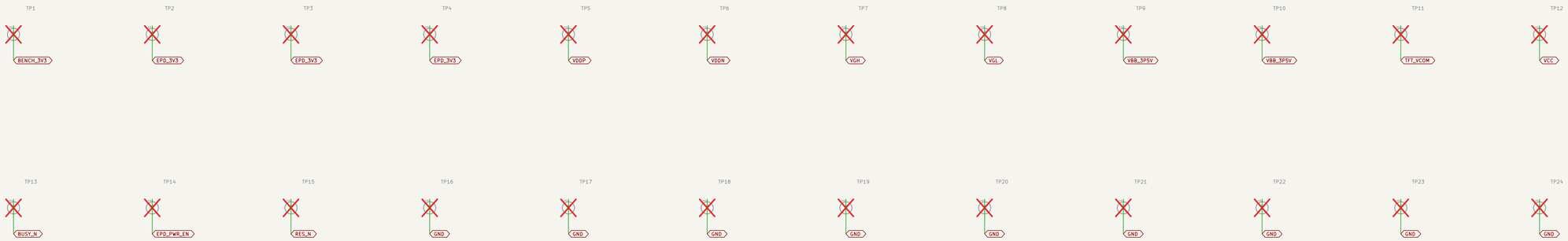
Direct 3.3V host interface — NO power-off signal isolation



Host first ON / last OFF. Screen OFF: no GPIO HIGH, no pull-ups.

J2.16 unused. No live plugging. Short cable <= 10cm.

Test pads: copper only, no parts to buy or solder



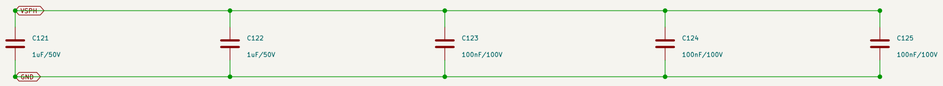
Sheet: /Host interface and test points/
File: interface.kicad_sch

Title: Interface — wired functional circuits

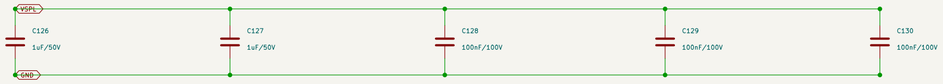
Size: User	Date: 2026-10-09	**Rev: 0.11-BASIC17**
KiCad E.D.A. 10.0.6		Id: 4/5

USER-SELECTED PROTOTYPE: biased capacitance / switching performance not verified.

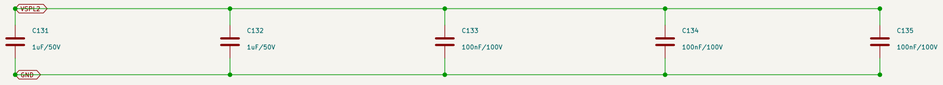
VSPH: C105 (1uF on panel page) + 2x1uF + 3x100nF = 3.3uF nominal



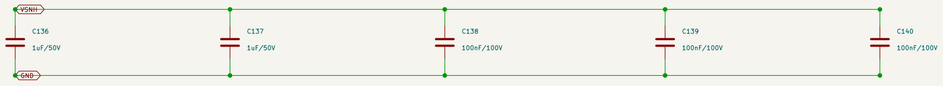
VSPL: C108 (1uF on panel page) + 2x1uF + 3x100nF = 3.3uF nominal



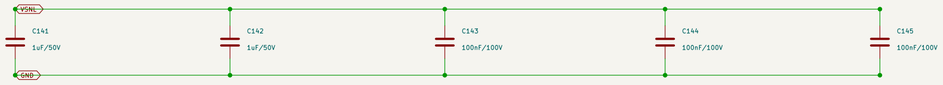
VSPL2: C111 (1uF on panel page) + 2x1uF + 3x100nF = 3.3uF nominal



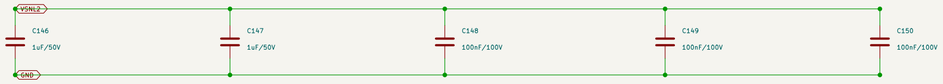
VSNH: C114 (1uF on panel page) + 2x1uF + 3x100nF = 3.3uF nominal



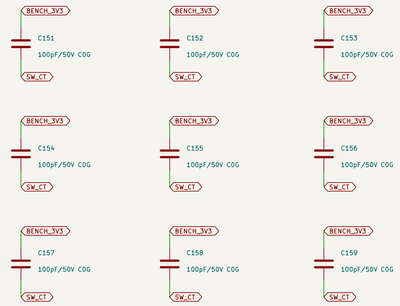
VSNL: C117 (1uF on panel page) + 2x1uF + 3x100nF = 3.3uF nominal



VSNL2: C120 (1uF on panel page) + 2x1uF + 3x100nF = 3.3uF nominal



CT: C36 + 9x100pF = 1nF COG



PUMP_P_LOW: C18(100nF) + 100nF + 22nF = 222nF



R18(15k) + R51(1k) = 16k

R8(100k) + R52(10k) = 110k



Sheet: /Basic-library composite additions/
File: basic-composites.kicad_sch

Title: Basic-library composites / prototype v0.11

Size: A2 | Date: 2026-10-11 | **Rev: 0.11-BASIC17**
KiCad E.D.A. 10.0.6 | Id: 5/5